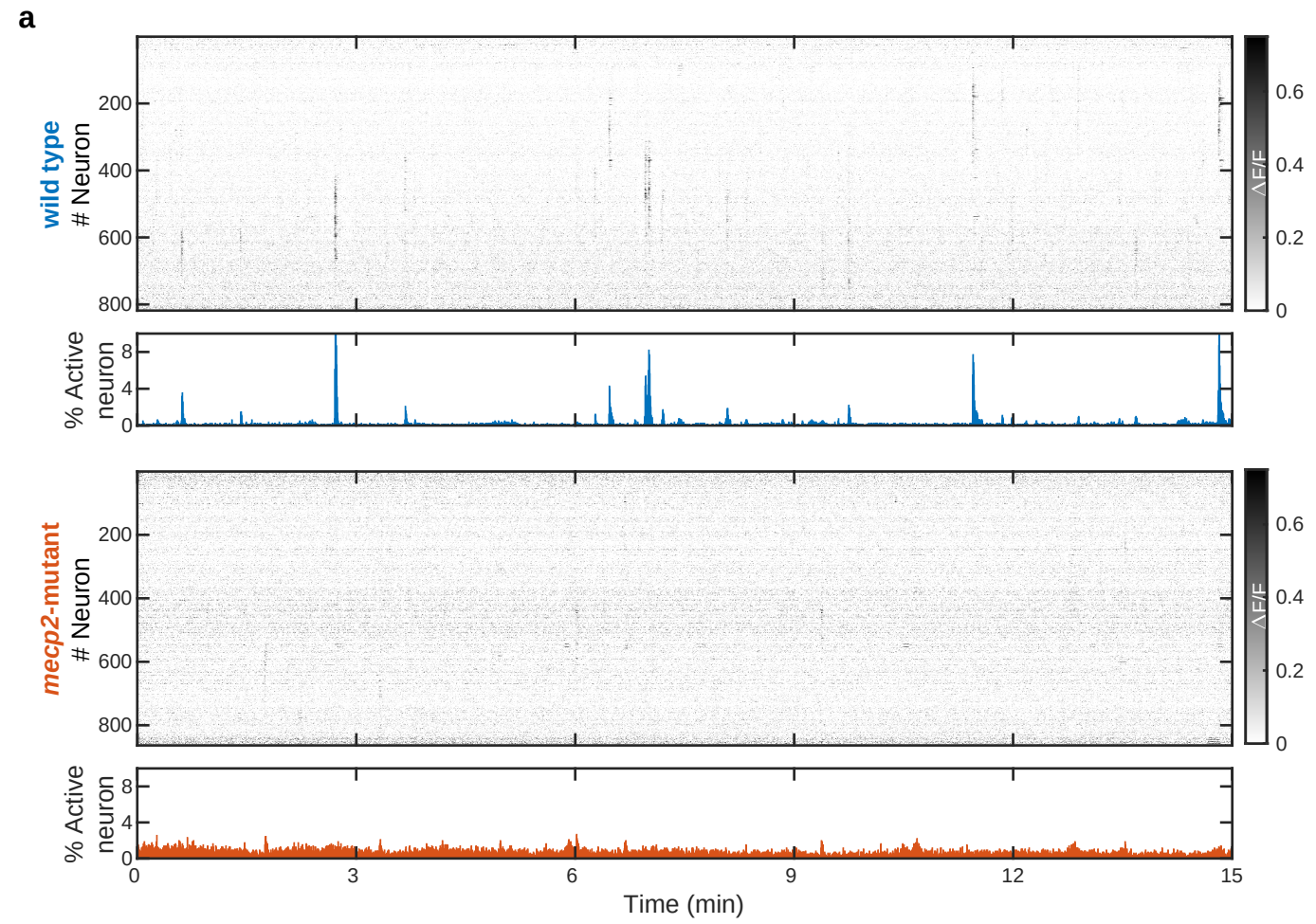
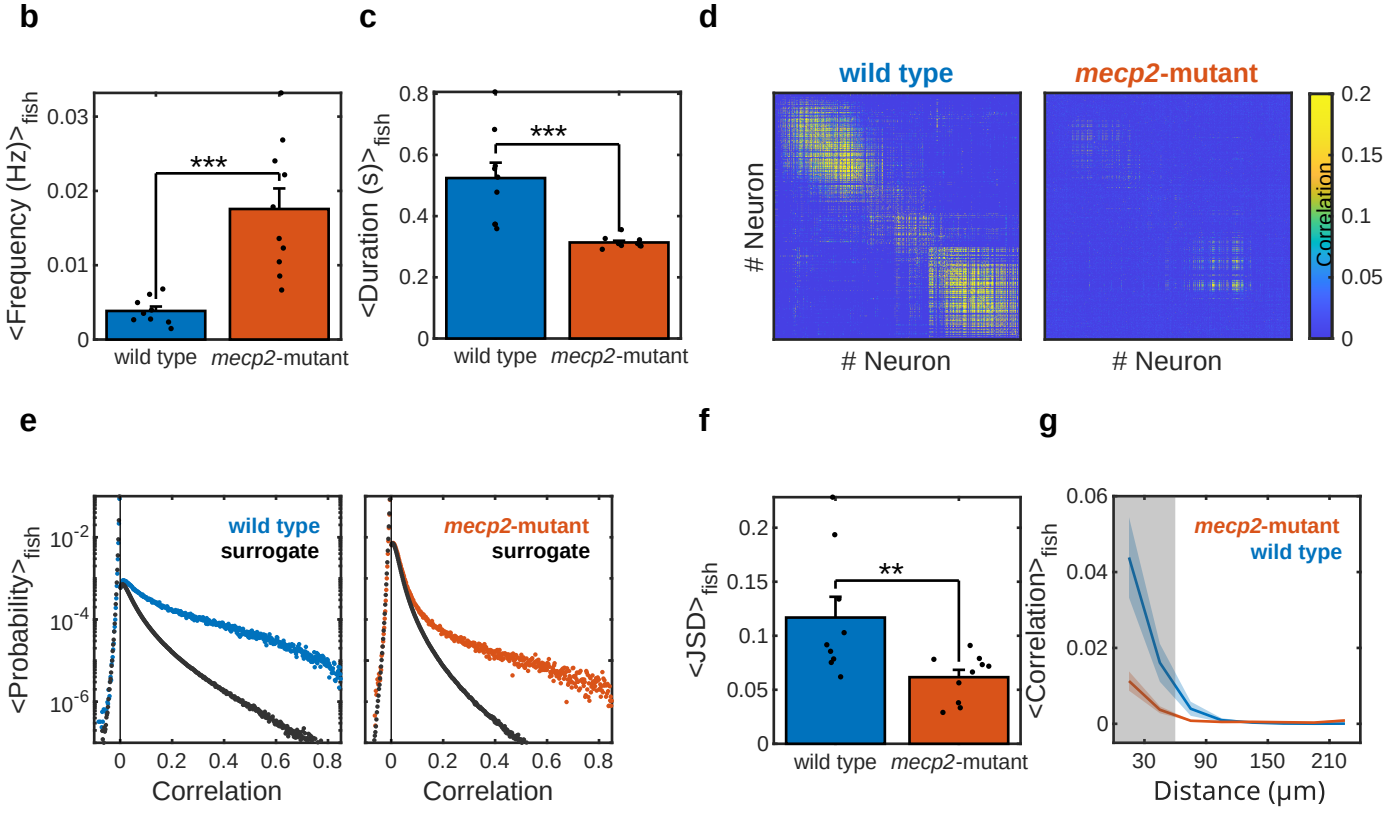

a
wild type
Neuron
200
400
600
800
ΔF/F
0.6
0.4
0.2
0
% Active neuron
8
4
0
mecp2-mutant
Neuron
200
400
600
800
ΔF/F
0.6
0.4
0.2
0
% Active neuron
8
4
0
0
3
6
9
12
15
Time (min)


b
<Frequency (Hz)>fish
0.03
0.02
0.01
0
***
wild type
mecp2-mutant
c
<Duration (s)>fish
0.8
0.6
0.4
0.2
0
***
wild type
mecp2-mutant
d
wild type
mecp2-mutant
Neuron
Neuron
Neuron
Correlation
0.2
0.15
0.1
0.05
0

e
<Probability>fish
10-2
10-4
10-6
wild type
surrogate
mecp2-mutant
surrogate
0
0.2
0.4
0.6
0.8
Correlation
Correlation
f
<JSD>fish
0.2
0.15
0.1
0.05
0
**
wild type
mecp2-mutant
g
<Correlation>fish
0.06
0.04
0.02
0
mecp2-mutant
wild type
30
90
150
210
Distance (μm)